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Amirul Ashraf

Curriculum Vitae

Introduction

Ashraf is a generalist with a deep streak in **performance optimisation**. From core contributions on Nethermind's flagship Ethereum client (cutting state/snap sync from ~3 hours to ~35 minutes, and shipping the HalfPath and FlatDB database layouts) to high-QPS backend work at Grab, Ashraf is most at home where there is a performance curve to bend. His work spans small contract jobs to enterprise employers such as Amazon and Grab, across C#/.Net, Go, Rust, Java, TypeScript and Ruby.

Skills

- **Performance optimisation**: profiling and hot-path tuning, sync and pipeline optimisation, memory and allocation tuning, benchmarking
- **Databases & storage**: RocksDB internals and tuning, custom storage layouts, MS-SQL, PostgreSQL, MongoDB, Solr; ZFS, NVMe-oF, iSCSI, RDMA
- **Blockchain**: Ethereum execution client internals, state/snap sync, Protocol Guild
- Proficient with server-side development using **C#/.Net**, **Go**, **Rust**, Java, Ruby on Rails
- Experienced in large micro-service architecture, high QPS, high reliability and big data systems; TypeScript, React and Electron on the frontend

Experience

Jun 2022 - Present

Ethereum Core Developer (.Net) (Contract), *Nethermind*, Remote, Kuala Lumpur, Malaysia

Core contributor on Nethermind's flagship Ethereum execution client. **Member of the Protocol Guild**.

- Worked on **The Merge** and subsequent protocol upgrades; Nethermind's network share grew from ~3% to ~33% over this time; it is now the second-largest Ethereum execution client.
- One of the **top three code contributors** since 2022, with **750+ merged PRs** (100+ performance, 100+ features) and 750+ PR reviews.
- Cut state/snap sync duration from ~3 hours to ~35 minutes; RocksDB tuning to cut sync write amplification and a receipts re-encoding that shrank that database by ~40%.

- Sped up **full pruning** on the original HashDB layout with a batched, key-ordered trie visitor, cutting pruning time 2–4× at a 4GB memory budget and 4–7× at 8GB; added low-priority and WAL-less writes so pruning no longer stalls block processing.
- Proposed and shipped the **HalfPath** database layout: ~40% faster block processing, 20× less database growth, 2× less sync write amplification, and enabled snap serving.
- Shipped the successor **FlatDB** layout, now Nethermind's **default state backend**: a further ~20% block processing speed-up on mainnet.
- Led a large-scale **architecture refactor** (dependency injection, backend-agnostic world state) to decouple the client from its state storage, making FlatDB possible and preparing the state layer for future backends such as a flat archive.
- Low-level performance work: intrusive change chains in the state journal, native-memory collections, deep-reorg OOM fix, batched RocksDB enumerations, custom Roslyn analyzers.

Aug 2020 - March 2022 **Software Development Engineer**, *Amazon Prime Video*, London, United Kingdom

Part of the Quality Platform SDK team for Digital Video, Mobile and Web Tech org.

- Developed tools and software to help write and automate testing.
- Reduced the service codebase by 60% by creating a DSL to remove boilerplate and replacing a significant portion of the service with an open-source library.
- Fixed bugs in internal tools, reducing the failure rate by 90% and improving start time through better observability, unit test coverage, and robustness against intermittent mobile device failures.

Dec 2018 - July 2020 **Software Engineer, Backend**, *Grab*, Petaling Jaya, Malaysia

One of the earliest engineers at the new KL development centre, assigned to the Safety team. Quickly adapted to Grab's vast backend architecture and helped bring the team up to speed. Promoted to Senior Software Engineer in April 2020.

- Implemented and maintained safety features such as customer face recognition integrations and textual **machine learning pipelines**.
- Designed and reviewed features with high bandwidth and reliability requirements; profiled and tuned hot paths under production load.

April 2018 - Nov 2018 **Senior Full Stack Developer (Contract)**, *Petronas ICT Sdn Bhd*, Kuala Lumpur

Designed and built an improved version of an existing gas monitoring system for Petronas Terengganu Operations Coordination Center.

- Improved an AngularJS frontend and a **.Net Web API** backend to requirements.
- Developed in parallel with a new gas pipeline simulation system from another team.

Nov 2017 - Dec 2017 **.Net Developer (Contract)**, *Petronas ICT Sdn Bhd*, Kuala Lumpur

Improved Bad Actor Management (BAM), an internal .Net system that tracks damaged or defective equipment across Petronas's infrastructure.

- Integrated with an existing IBM BPM system as a transitional solution.
- Integrated with an SAP system through SOAP.

Nov 2014 - Oct 2017 **Full Stack Developer (Part Time)**, *NetByteSEC*, Bangi

- Developed DEALS, a website defacement tracker, including a backup URL scraper service in the **Phoenix framework**; practised peer review across the team.
- Developed a Quality Management System for a client in Ruby on Rails, with **integration and feature tests in RSpec and Fabricator** and a Docker setup for easier deployment.

Education

Aug 2025 **Master of Artificial Intelligence (Coursework)**, *University of Malaya*
Pass with Distinction, CGPA: **3.74**. A+ for the research project.
Research project — **Autonomous driving**: *Improving Reliability of Autonomous Driving Agent by Unifying Feature Vector from Multiple Viewpoints.*

Oct 2017 **Bachelor of Science, Computer Science**, *International Islamic University Malaysia (IIUM)*
CGPA: **3.601**. Awarded "Best Computer Science Student" of the batch.

Personal Projects

2026 **TreeTerm**, *TypeScript, Electron*, github.com/asdacap/treeterm
A terminal multiplexer for running multiple AI coding agents across machines over SSH. A per-host daemon keeps sessions alive across client restarts and disconnects; workspaces are Git worktrees; an optional auto-approve mode uses a separate model to judge whether a pending action is safe.

2026 **Parrot**, *.Net Native AOT*, github.com/asdacap/parrotdotnet
A local coding agent that runs interactively, takes one-shot prompts, or is hosted as a service for remote clients. Supports configurable model providers with cost-aware model tiers, persistent sessions, and delegated child agents with deliberately bounded scope and context.

2026 **GreenDotRDMA**, *Rust*, github.com/asdacap/greendotrdma
A storage-appliance web UI exporting ZFS zvols over NVMe-oF and iSCSI, showing at a glance whether each export is actually served over RDMA, with LVM, partitioning and scheduled snapshots.

Activities

2014-2019 **ACM-ICPC**: problem setter for the Asia Regional Kuala Lumpur Contest 2019 and the Al-Khawarizmi National Programming Competition 2018–2019; top 60 of 150 Southeast Asian teams at the Asia Regional Singapore 2015.